

Exercise 1

Consider a vector:

What is the value of:

`c(4,6,5,7,10,9,4,15) < 7`

- a. TRUE, FALSE, TRUE, FALSE, FALSE, FALSE, TRUE, FALSE
- b. TRUE, TRUE, TRUE, FALSE, FALSE, FALSE, TRUE, FALSE
- c. FALSE, TRUE, TRUE, FALSE, FALSE, FALSE, TRUE, FALSE
- d. TRUE, TRUE, TRUE, TRUE, TRUE, FALSE, TRUE, FALSE
- e. TRUE, TRUE, TRUE, FALSE, FALSE, FALSE, TRUE, FALSE

Exercise 2

Consider two vectors:

`p <- c(3, 5, 6, 8)`

and

`q <- c(3, 3, 3)`

What is the value of:

`p+q`

- a. 6, 8, 6, 8
- b. 6, 8, 0, 0
- c. 6, 8, NA, NA
- d. 3, 5, 6, 8 Warning message: In `p+q` : longer object length is not a multiple of shorter object length
- e. 6, 8, 9, 11

Exercise 3

Consider two vectors

`a = c(1,3,4,7,10,0)`

`b = c(1,2)`

what is the value of `a+b` ?

Exercise 4

If

```
z <- 0:9
```

then what is the output from the following R-statements:

```
digits <- as.character(z)
```

```
as.integer(digits)
```

- a. Error in subset. object 'z' not found
- b. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- c. "NA", "NA", "NA", "NA", "NA", "NA", "NA", "NA", "NA"
- d. "0", "1", "2", "3", "4", "5", "6", "7", "8", "9"
- e. 0, 0, 0, 0, 0, 0, 0, 0, 0

Exercise 5

Consider the vector:

```
x <- c(1,2,3,4)
```

What is the value of k for:

```
(x+2)[(is.na(x)) & x > 0] -> k
```

- a. 1, 2, 3, 4
- b. 1, 4, 9, 16
- c. Error: object 'k' not found
- d. 3, 4, 5, 6
- e. numeric(0)

Exercise 6

Consider the following vectors:

```
s = c("a","b","c","d","e")
```

```
t = c("f","g","h","i","j")
```

What is the value of s+3 ?

What command would you use to combine them into a single vector in alphabetical order?

Exercise 7

Consider the following vectors:

$s = c("a", "b", "c", "d", "e")$

$v = 1:5$

if:

$z = c(s, v)$

What is the value of $z[5:10]$?

- a. 1,2,3,4,5
- b. "a", "b", "c", "d", "e"
- c. "1", "2", "3", "4", "5"
- d. "e", "1", "2", "3", "4"